AEGIS PROTOCOL: THE ONLINE DISK DEFENSE

Tryst Mathematical Modelling Challenge (TMMC)

Structure:

- A provided set of satellites
- Online sequence of disks
- Choose the most optimum satellite to activate/use
- Objective: Minimize the hitting set

This is an Online Geometric Hitting Set problem

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THE BASELINE STRATEGY

The first basic step that came to us was not directly about "how do we pick the optimal satellite?" but "how do we even search efficiently?"

The **step 1** was to realise that **Brute force won't work**

Explaination:

- $N \leq 10^5$ satellites
- $Q \leq 10^4$ anomalies
- Naively checking every satellite for every disk:

O(NQ)=10^9

Further in **step 2** we observed the radius<=100 constarint

And, we collectively agreed upon working on a **local spatial neighborhood**, and running teh disc queries locally in the neghbouring grids.

Step 3: The final conclusion: precomputation techniques

We concluded upon using grid hashing, wherein:
1) Dividing the whole plan online into 100x100 subgrids
2) Each satellite mapped to a cell
3) The new Average Query Time=O(1): Now, instead of scanning all satellites, we need to access only spatially relevant regions.

THREE COMPETING HEURISTICS: OUR INTERNAL DEBATE

Once spatial search was solved, the real question remained:
If forced to activate a new satellite, which one should we choose? This question raised a debate amongst us.
So, we decided to test all of them.

Krishna: Sparse-First Philosophy

Shlesha: High-Density Optimization

Aayush: Geometric Centrality

Test Case	Description	N	Q	M1 Activated	M1 Time(s)	M2 Activated	M2 Time(s)	M3 Activated
EC10_overlapping_blobs	*5 blobs spaced 150 units apart (radius≈100). High overlap zones.*	5000	500	**58**	**1.0643**	71	1.1436	77
EC1_ultra_sparse	*50 sats, huge ±1e6 space. Each disk barely reaches 1 sat.*	50	500	**50**	**0.0013**	**50**	**0.0012**	**50**
EC2_ultra_dense	*5000 sats packed in 200×200 box. Every disk has ~100s of candidates.*	5000	500	**54**	**3.827**	66	3.8047	69
EC3_single_tight_cluster	*All 5000 sats in 10×10 patch; disks from across ±1e6.*	5000	500	**13**	**3.995**	**13**	**4.0804**	**13**
EC4_many_micro_clusters	*100 clusters of 50 pts each. Disks sweep cluster centers repeatedly.*	5000	1000	282	0.0662	**268**	**0.0607**	425
EC5_line_sweep	*1000 sats on y=0 line. Disks sweep along the line.*	1000	500	**372**	**0.0032**	**372**	**0.003**	**372**
EC6_concentric_rings	*Sats on rings r=200→100k. Disks hit random ring arcs.*	2800	500	**333**	**0.0096**	**333**	**0.0094**	**333**
EC7_skewed_density	*90% of sats in tiny 50×50 patch + sparse field. Mixed disks.*	5000	500	**148**	**3.1808**	**148**	**3.2038**	**148**
EC8_grid_with_holes	*Grid sats only in Q1+Q3 diagonal. Disks hit all 4 quadrants.*	2500	500	**260**	**0.007**	**260**	**0.0072**	**260**
EC9_hot_cold_clusters	*20 clusters; 4 'hot' clusters get 50% of disk traffic.*	4000	1000	**82**	**0.1467**	86	0.1574	184

KRISHNA'S APPROACH: LOW DENSITY OPTIMIZATION

I calculated the density for each satellite (the number of satellites within its cell and the 8 surrounding cells) and selected the satellite with the minimum density.
Logic: If a point is isolated, it's the only way to cover that specific area. Activating it now saves us from being forced to activate it later when a disk perfectly targets that empty space

- This algorithm has **O(1)** time complexity **per anomaly**.
- But, the result of our experiment did not support the rationale. This is because the **probabilistic argument** of a anomaly being in a the region of a satellite with less neighbours is incorrect.
- I finally understood that since each anomaly covers a satellite, it stands to have higher probability to be located around high density areas.

SHLESHA'S APPROACH: HIGH-DENSITY HEURISTIC

So my basic intitution went on like, the region with the highest density will have the maximum probability of having an anomaly around it and therefore active points and hence we must prefer such neighbouring regions in every iteration. We can go on to have overlap regions and use the concept of reusing the previously identified active points. (This was in fact verified by Krishna's analysis later on)

1 Mathematical formulation

$$p^* = \arg p \in D \max \text{density}(p)$$

2 Time complexity

Grid Construction (Pre-computation): O(N)
Density Computation (Each point checks neighbouring cells): O(N)
Query Processing: O(Q) [O(1) Per query]
TOTAL: O(N+Q)

3 Conclusion

The high-density strategy balances computational efficiency and geometric reuse potential, providing a practical near-optimal online approximation under spatial uncertainty. The only factor that this method ignores is the size of the anomalies, which could impact net accuracy.

AAYUSH GEOMETRIC CENTRALITY

- For the given k (let) points, choose a point p such that the distance from the centre is **minimum**.
- A straightford algorithm with the k satellites, which, in comparison to n, has complexity **O(1) per query**.
- I argued that if we were to study the (size, location, and probability) of the anomalies, then, by geometrical constraints and symmetry, we would observe larger anomalies in the centre and smaller ones towards the ends.
- So, an **active satellite** could capture a larger geometric area if it were located in the central regions. A larger area captured implies that all future discs that could appear (with high probability) in that area are satisfied.

- The counterargument by others was that the probability of a disc appearing is not a function of location but of density around the point (as any disc must cover some point). They really convinced me of the inefficiency of this method.
- But surprisingly, this metric performed well compared to others.
- On second thought, we realised that the selected grid is very vast. So, for general test cases with a limited number of satellites and queries, the density of each satellite could be very sparse and have little impact. So, geometric interpretation matters.
- However, in a few cases Shlesha's approach did overpower

COMPILATION: THE SCORE

- When using custom test cases, Aayush seems to be the victor: the least-distance-to-the-centre metric proves optimal. It optimises the size of the anomalies.
- This is because, in a highly random system, the immediate density (local) is very low. And this metric takes dominance.
- But this metric fails in some test cases, which can be compensated for by other metrics derived from Shlesha's high-density optimisation method.
- So, we decide upon new metrics, normalise them and use empirical weights (as per their importance) to get:

SCORE = 0.5(normalized distance) + 0.3(normalized active density) + 0.2(normalized density)

normalised distance = 1.0 - (distance/radius)^2 [Geometric interpretation, optimise size]

normalised density = (density/maximum density) [Probabilistic interpretation, optimise location]

normalized active density = 1.0/(1.0 + active density) [A metric to regulate and avoid already active regions]

CONCLUSION

- To parse the inputs and optimise search, we applied pre-computational techniques. We constructed a static hash table to record the number of satellites per subgrid. In parallel, we calculate, for each satellite, the subgrid it belongs to and its associated density for further analysis.
- Then we take the input for each query (each anomaly) and check if some active satellite already exists within it. If it does, we select it trivially and proceed further. If none do, we analyse each satellite with the optimised SCORE Formula.
- We select the satellite with the highest score, and update the active satellites in a dynamic table for each subgrid.
- TIme complexity (similar as each arugment before) (Note finding satellites inside an anomaly takes insignificant time due to pre-computation):
  - Grid Construction (Pre-computation): O(N)
  - Density Computation (Each point checks neighbouring cells): O(N)
  - Query Processing: O(Q) [O(1) Per query]
  - TOTAL: O(N+Q)

THANK YOU!

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